BLACK SWAN · MVO4 PARAMETRIC SWEEP

Signature Sweep Report

Full **K × Seeds × MinAppear × w1** grid across Swan 8, Swan 9, Swan 10D and Swan 11A, with crash timing from the **Black Swan Signatures** workbook and **max_sharpe locked at rf = 0.0**. September 2026.

01 Parameter ranges

Every combination was run on all four swans.

K (CLUSTERS)	SEEDS (DBI ENSEMBLE)	MINAPPEAR (VOTE THRESHOLD)	W1 (PER-STOCK CAP)
5, 6, 7, 8, 9, 10, 11, 12	3, 5, 7, 9, 11, 13	2, 3, 4, 5	0.05 – 0.17, step 0.02

5,376	4,130	871	4
RUNS	SOLVED (76.8%)	COMBOS SOLVED ON ALL 4 SWANS	SWAN 8, SWAN 9, SWAN 10D, SWAN 11A

The 1,246 unsolved runs are 756 where `max_sharpe` had no feasible solution (mostly small baskets at low w1, where the stock count times w1 is under 1) and 490 where MinAppear was higher than Seeds, so no stock could collect enough votes.

02 Standing methodology

max_sharpe locked at rf = 0.0

Every combination solves `max_sharpe(rf=0.0)` directly. There is no retry at rf = 0.01 and no fallback to minimum volatility. A combination that can't solve at rf = 0.0 is recorded as failed.

03 What changed in this version

Three changes from the Locked Sweep Report. All three affect the crash timing and the training window; the grid, preselection and optimizer are unchanged.

CHANGE	LOCKED SWEEP REPORT	THIS VERSION
Crash timing T1	Weighted by beta correlation (the notebook's executed line)	Plain average of prior swans, as in the Signatures workbook, rows 69–75
Swan 10D crash timing	T1 195, T2 415	T1 196, T2 297 (Signatures workbook). Swan 11A added: T1 34, T2 54
Training window	Set by hand per swan	Ends the Friday before the swan's week and starts 13.5 months earlier. Moved back only if the window can't hold T1 + T2 trading days

SWAN	TRAINING WINDOW	TRADING DAYS	T1 / T2	PREVIOUS WINDOW	PR
Swan 8	2017-08-07 → 2018-09-28	290	74 / 216	2017-07-22 → 2018-09-28	
Swan 9	2018-12-30 → 2020-02-14	284	72 / 200	2019-01-01 → 2020-02-15	
Swan 10D	2020-11-15 → 2021-12-31	284	66 / 188	2020-11-16 → 2021-12-31	
Swan 11A	2023-12-30 → 2025-02-14	282	79 / 198	2023-10-31 → 2025-01-31	

Swan 8's 13.5-month window starts 2017-08-13 and holds 285 trading days, 5 short of its 290-day crash curve, so it starts 5 trading days earlier. The other three fit with 12, 30 and 5 days to spare. Swan 11A no longer needs a longer window: its previous 458-day window came from the weighted T1 and the old Swan 10D T2.

04 Best result

This combination ties the highest worst-case PR in the grid (13 combinations tie at 44.6%), and it has the highest mean PR of all of them.

K=7 · Seeds=3 · MinAppear=3 · w1=0.17 BEST WORST CASE · BEST MEAN AMONG TIES

61.6%	44.6%
MEAN PR	MIN PR · SWAN 10D

Swan 8	53.1% 15→7	Swan 9	49.1% 20→7	Swan 10D	44.6% 14→7
AMZN	17.6%	AMZN	17.3%	MCD	17.5%
KO	17.6%	COST	17.3%	VZ	17.5%
NEE	17.6%	MDLZ	17.3%	MRK	17.1%
UNH	17.6%	MRK	17.3%	MDLZ	16.8%
DUK	11.8%	WMT	17.3%	SBUX	12.6%
MO	10.4%	LLY	7.1%	AMT	9.4%
EXC	7.6%	MO	6.4%	DUK	9.1%

Swan 11A	99.5% 13→7
DUK	17.0%
EXC	17.0%
KO	17.0%
PG	17.0%
VZ	17.0%
PEP	11.3%
MDLZ	3.7%

"15→7" means 15 stocks were preselected into MVO and the optimizer weighted 7. The same combination under the Locked Sweep Report settings scored 57.4% / 48.8% / 45.2% / 98.2% (mean 62.4%, worst case 45.2%). The highest mean PR in the grid is K=12 / Seeds=3 / MinAppear=3 / w1=0.11 at 65.1%, but its worst case is 33.9% on Swan 9.

05 Swan 9 data check

TREAT THE WORST-CASE RESULTS AS PROVISIONAL

The expected return the optimizer uses (`R_fwd_30_MA` , built from analyst price targets collected the Friday before each swan) looks unreliable for Swan 9. Across stocks, Swan 9's targets rank the COVID crash's winners and losers almost perfectly: a rank correlation of +0.77 with each stock's return over the next 30 days, against +0.41 for past price momentum. Data available on February 14, 2020 shouldn't be able to do that, so these targets were likely collected after the crash. Swan 10D's and 11A's targets match what look like genuine point-in-time snapshots, and Swan 8's show no comparable correlation.

Point-in-time targets are being rebuilt in a new optdata database. Until Swan 9's are verified, treat its results, and the worst-case rankings that depend on them, as provisional.

06 rf = 0.0 vs. the old fallback chain

The best result from section 04, re-solved with the previous chain (try rf = 0.01, then rf = 0.0, then minimum volatility).

RF FALLBACK CHAIN	
Swan 8	**53.1%**
Swan 9	**49.1%**
Swan 10D	**44.6%**
Swan 11A	**98.4%**
Mean / worst	**61.3% / 44.6%**

RF = 0.0 ONLY (STANDARD)	
Swan 8	**53.1%**
Swan 9	**49.1%**
Swan 10D	**44.6%**
Swan 11A	**99.5%**
Mean / worst	**61.6% / 44.6%**

In this version the lock changes only Swan 11A, where the fallback chain solved at rf = 0.01 and gave a slightly different basket. For this combination the effect is small: 0.3 points on the mean, none on the worst case.

07 Top combos by mean PR

The small numbers under each PR are stocks preselected into MVO → stocks weighted.

K	SEEDS	MINAPPEAR	W1	SWAN 8	SWAN 9	SWAN 10D	SWAN 11A	MEAN	MIN
12	3	3	0.11	85.1% 10→10	33.9% 14→9	44.7% 14→9	96.9% 12→9	**65.1%**	33.9%
12	3	3	0.13	83.0% 10→8	30.9% 14→8	42.5% 14→8	99.7% 12→8	64.0%	30.9%
12	5	4	0.11	85.1% 10→10	29.0% 15→10	44.7% 14→9	96.9% 12→9	63.9%	29.0%
12	3	3	0.17	82.3% 10→7	29.0% 14→7	44.6% 14→7	99.5% 12→7	63.9%	29.0%
12	3	3	0.15	81.3% 10→7	30.2% 14→8	44.0% 14→8	99.8% 12→7	63.8%	30.2%

K	SEEDS	MINAPPEAR	W1	SWAN 8	SWAN 9	SWAN 10D	SWAN 11A	MEAN	MIN
12	5	4	0.13	83.0% 10→8	27.1% 15→8	42.5% 14→8	99.7% 12→8	63.1%	27.1%
11	3	3	0.13	69.8% 9→8	30.9% 14→8	51.6% 13→8	99.7% 12→8	63.0%	30.9%
11	5	5	0.13	69.8% 9→8	30.9% 14→8	51.6% 13→8	99.7% 12→8	63.0%	30.9%
12	5	4	0.17	82.3% 10→7	24.9% 15→7	44.6% 14→7	99.5% 12→7	62.8%	24.9%
7	3	3	0.09	71.5% 15→11	32.5% 20→12	49.6% 14→12	97.5% 13→11	62.8%	32.5%

08 Top combos by worst-case PR

K	SEEDS	MINAPPEAR	W1	SWAN 8	SWAN 9	SWAN 10D	SWAN 11A	MEAN	MIN
7	3	3	0.17	53.1% 15→7	49.1% 20→7	44.6% 14→7	99.5% 13→7	**61.6%**	**44.6%**
6	3	2	0.17	47.6% 19→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**
6	5	2	0.17	47.6% 19→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**
6	5	3	0.17	47.6% 19→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**
6	5	4	0.17	47.6% 18→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**
6	7	2	0.17	47.6% 19→9	49.1% 20→7	44.6% 15→7	99.5% 13→7	60.2%	**44.6%**
6	7	3	0.17	47.6% 19→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**
6	7	4	0.17	47.6% 19→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**
6	7	5	0.17	47.6% 19→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**
6	9	4	0.17	47.6% 19→9	49.1% 20→7	44.6% 14→7	99.5% 13→7	60.2%	**44.6%**

Every leading K=6 / w1=0.17 combination holds the same baskets on Swan 9, Swan 10D and Swan 11A; they differ only on Swan 8. The K=7 basket ties them on the worst case and beats them on Swan 8, and so on the mean. The worst case in every row is Swan 10D.

09 Average PR by parameter value

Mean PR averaged over every complete combination with that value.
Longer bar is better.

K (clusters)



Seeds (DBI ensemble size)



MinAppear (vote threshold)



w1 (per-stock weight cap)



w1=0.15	52.5%
w1=0.13	52.4%
w1=0.07	52.1%
w1=0.05	51.5%

AVERAGES AND EXTREMES DISAGREE ON W1

On average, w1 = 0.09 does best. But every combination at the top of the worst-case table uses w1 = 0.17. A higher cap lets the optimizer concentrate in the few names that protect best on the hardest swan, while a lower cap gives steadier results in the typical case. Note that only 6 combinations with w1 = 0.05 solved on all four swans.

10 Notes

- **All results are in-sample.** The best combinations are chosen by looking at all four swans' outcomes. Treat them as a description of the parameter space, not a forecast.
- **Weights can end slightly above w1.** Holdings under 3% are dropped and the rest renormalized after the cap is applied, so some weights exceed the cap by up to about a point (e.g. MCD and VZ at 17.5% on Swan 10D in section 04). This is how the original notebook works.
- **Swan 11A's forward-return snapshot is dated 2025-01-31**, two weeks before its new window end of 2025-02-14. The other three snapshots match their window ends.
- **Reproducibility.** Produced with the `mvo4_sweep` folder (`results/signature_avg_13p5mo`). A parity check confirms the sweep code matches the original `mvo4.ipynb` logic step by step on all four swans. Run on Python 3.13, scikit-learn 1.7.2, PyPortfolioOpt 1.6.0.

Full results, holdings and marginal tables: `mvo4_sweep/results/signature_avg_13p5mo/summary.xlsx`. The Locked Sweep Report settings are kept as `presets/locked_sweep_report.py` and reproduce that report exactly.